

BSE 4105 — Software Integration & Deployment

Exercise 06: CI/CD with GitLab

Yohana Mahamat Abdelrassoul | yohana.mahamatabdelrassoul

30 September 2026

1. Introduction

This report describes a complete GitLab CI/CD pipeline for a Flask application that stores user records in PostgreSQL. The work was done in the personal class project, on a dedicated **develop** branch (not master/main). The pipeline builds a versioned Docker image, runs pytest against a disposable database, and deploys the stack to the shared class host with Docker Compose. The live application is reachable on the student host port assigned in the Compose lab (30847).

2. Required links and repository details

Item	Value
Application home	http://ldapgit.craneccloud.io:30847/
Users API	http://ldapgit.craneccloud.io:30847/users
GitLab project	https://gitlab.craneccloud.io/2026-bse-4105/yohana.mahamatabdelrassoul
CI/CD branch	develop (default branch remains main; no CI/CD work on master/main)
Pipelines	https://gitlab.craneccloud.io/2026-bse-4105/yohana.mahamatabdelrassoul/-/pipelines
Latest commit	8029ff4 — Complete GitLab CI/CD pipeline with Docker Hub images.
Docker Hub image	abdeldev235/cicd-app:0.2 (also tagged latest)

3. Application and Docker setup

The service is a small Flask API. `/` returns a greeting that identifies the student account. `/users` queries a PostgreSQL database named **user** (single table of user details, seeded from `db_init/init.sql`). Compose service names are **yohana-web** and **yohana-db** so they do not collide with classmates on the shared Docker host. Host ports are **30847**→**5000** (web) and **31847**→**5432** (Postgres). Resource limits match the lab template (web 0.50 CPU / 256 MB; database 0.25 CPU / 128 MB). The web service runs the versioned Hub image rather than bind-mounting source, so deploy pulls an artefact that already passed tests.

4. Docker Hub images

Images are pushed to Docker Hub as `abdeldev235/cicd-app:0.2` and `abdeldev235/cicd-app:latest`. GitLab CI logs in with **DOCKERHUB_USERNAME** and **DOCKERHUB_TOKEN** so pull/push is authenticated. A private Hub repository was requested, but the Hub account plan returns *no more private repositories available*; the repository therefore remains authenticated Docker Hub (login required to push; tags 0.1/0.2 as specified). The running container on the class server is `abdeldev235/cicd-app:0.2`.

5. GitLab CI/CD pipeline

The pipeline is defined in `.gitlab-ci.yml` at the repository root and is restricted to **develop** with job rules. Stages:

Build. Docker 24 client on the class GitLab runner. Login to Docker Hub, build `./app`, tag 0.2 and latest, push both tags.

Test. Creates a uniquely named network and Postgres container (names include `$CI_JOB_ID` so jobs do not stop classmates' containers). Waits for `pg_isready`, loads `init.sql`, starts the Flask image with `POSTGRES_HOST=db`, runs `pytest` (hello route, `/users` list, database connectivity), then cleans up.

Deploy. Alpine job installs OpenSSH, loads the **SSH_PRIVATE_KEY** file variable (ECDSA), and SSHes to **DEPLOY_HOST** as **DEPLOY_USER**. The server logs into Docker Hub, pulls tag 0.2, and runs `docker compose up -d` in `$DEPLOY_HOME/microservices/yohana.mahamatabdelrassoul`. The ECDSA public key is in `~/.ssh/authorized_keys` on `ldapgit.craneccloud.io`.

6. CI/CD variables

Group variables supply Postgres credentials (**POSTGRES_USER**, **POSTGRES_PASSWORD**, **POSTGRES_HOST**, **POSTGRES_PORT**). Project variables used for deploy: **DEPLOY_HOST** (ldapgit.cranecloud.io), **DEPLOY_USER** (class SSH username), **DEPLOY_HOME** (home directory), **SSH_PRIVATE_KEY** (File type), **DOCKERHUB_USERNAME**, **DOCKERHUB_TOKEN**, and project **POSTGRES_DB=user** so tests match the assignment database name. Note: if **DEPLOY_HOST** is marked Protected, it is only injected on protected branches; **develop** should be protected (or the variable unprotected) so the deploy job receives the hostname.

7. Verification

After image push and Compose up, */* returns HTTP 200 with the student greeting. After PostgreSQL finished initdb, */users* returns JSON for ten seeded students. Compose project name on the host is yohanamahamatabdelrassoul with containers yohana-web and yohana-db. The same README.md in the GitLab **develop** branch documents these links for the MUELE submission.

8. Conclusion

The exercise delivers a three-stage GitLab pipeline, versioned Docker Hub images, and a running Flask + Postgres deployment on the class VM at port 30847. Work stayed on **develop**, used unique test/deploy names, and reused the assigned host ports so other students' services on ldapgit.cranecloud.io were not disturbed.